

Summary

Electronics and Communication Engineering student with strong interests in software engineering, analog IC design, and developer tools. Experienced in building desktop applications using modern C++, designing CMOS analog circuits in Cadence Virtuoso, and developing engineering utilities in Python. Seeking opportunities to apply both software and electronics knowledge to real-world engineering challenges.

Skills, Tools and Languages

Skills	Tools	Languages
• Circuit Design • Communication System • PCB Design • Web development • Python Development • C++ Development	• VS code • Visual Studio • JetBrains Rider • JetBrains WebStorm	• C/C++ • Python • HTML, CSS • Javascript • React JS • Tailwind CSS

Projects

1. CMOS Instrumentation Amplifier Design using Cadence Virtuoso

Designed and implemented a two-stage CMOS operational amplifier using 90 nm CMOS technology in Cadence Virtuoso. Building upon the validated op-amp design, developed a CMOS instrumentation amplifier optimized for analog signal conditioning applications.

- Designed and simulated a two-stage CMOS operational amplifier using 90 nm technology in Cadence Virtuoso.
- Developed a CMOS instrumentation amplifier by integrating the designed operational amplifier stages.
- Performed transistor sizing, DC, AC, transient, and stability analyses to validate amplifier performance.
- Optimized the design for gain, bandwidth, stability, and power efficiency through iterative simulation.
- Utilized Cadence Virtuoso for schematic design, simulation, and performance verification.
- Research paper based on this work has been accepted for presentation at the ICICET 2026 Conference and is pending publication.

2. PDF Graph Digitizer & Bode Plot Analyzer | Python

[github-link](#)

Developed a Python-based desktop application to digitize and replot CMOS amplifier simulation graphs from PDF reports, addressing the lack of suitable graph extraction tools during analog IC design research.

- Built an interactive GUI to import PDF graphs, calibrate axes, and accurately extract multiple curves.
- Automated the conversion of simulation plots into digital datasets for further analysis and comparison.
- Implemented Bode plot analysis to calculate Gain Margin and Phase Margin from extracted frequency response data.
- Enabled export of processed data to CSV, significantly reducing the time required for analyzing and reproducing Cadence Virtuoso simulation results.

- Developed to support CMOS amplifier characterization and research workflows.

3. code ZEN – Lightweight Code Editor & Developer Tool | C++, OpenGL, GLFW, Dear ImGui [github-link](#)

Designed and developed a lightweight code editor for Windows after identifying the high resource usage of traditional IDEs while developing Unreal Engine projects. Focused on delivering a responsive development environment with essential coding and terminal features.

- Developed a modular desktop application in modern C++ using OpenGL, GLFW, and Dear ImGui Docking.
- Implemented a multi-panel workspace with docking, persistent layouts, and asynchronous file indexing for improved responsiveness.
- Integrated a native Windows ConPTY terminal supporting multiple tabs, split views, scrollbar, clipboard operations, and Git command shortcuts.
- Built a configurable code editor with customizable compile-and-run workflows for multiple programming languages.
- Designed a real-time output console with color-coded compiler and runtime logs to simplify debugging.
- Structured the application using CMake, multithreading, and modular components to support future feature expansion.

4. Personal Portfolio Website | React.js, Vite, Tailwind CSS [portfolio-Link](#)

Designed and developed a responsive personal portfolio website to showcase software development, game development, and electronics engineering projects.

- Built a modern single-page application using React.js, Vite, and Tailwind CSS with a responsive and mobile-friendly interface.
- Developed reusable React components to present projects, technical skills, experience, and contact information.
- Implemented smooth navigation, optimized asset loading, and fast build performance using Vite.
- Designed the website as a centralized platform to demonstrate projects, technical expertise, and professional achievements.

Internship

Eastern Railway S & T Department

Kavach System: Safety Integration

- Studied the **Kavach** Automatic Train Protection (ATP) system and its role in enhancing railway safety and preventing collisions.
- Gained an understanding of railway signaling, train communication, and safety integration concepts used in modern railway operations.
- Observed the implementation and operation of signaling and telecommunication equipment in a safety-critical railway environment.

Education

Ramkrishna Mahato Government Engineering College, Purulia

2022 – Present

Bachelor of Technology — Electronics and Communication Engineering

Class XII

WBCHSE --- Ramjibanpur Babulal Institution

2021

Class X

WBBSE --- Ramjibanpur Babulal Institution

2019